

Assingment—3

Question 1: Zero-Shot Prompting (Palindrome Number Program)

Prompt: generate python code to check a number is palindrome or not

Code:

```
def is_palindrome(num):  
    # Convert the number to string  
    str_num = str(num)  
  
    # Check if the string is equal to its reverse  
    return str_num == str_num[::-1]  
  
## Example usage  
number = 121  
if is_palindrome(number):  
    print(f"{number} is a palindrome.")  
else:  
    print(f"{number} is not a palindrome.")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\3.py"  
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\3.py"  
121 is a palindrome.  
PS C:\Users\prath\.vscode\github> █
```

Question 2: One-Shot Prompting (Factorial Calculation)

Prompt: Write a Python function to find the factorial of **5**. Please use a simple **while loop** and print each step of the calculation so I can see how the number multiplies.

Factorial of 5= (5*4*3*2*1=120)

Code:

```
def factorial(n):  
    result = 1  
  
    i = 1  
  
    print(f"Calculating factorial of {n}:")  
  
    while i <= n:  
        result *= i  
  
        print(f"Step {i}: {result}")  
  
        i += 1  
  
    return result  
  
# Example usage  
  
number = 5  
  
factorial(number)
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
Calculating factorial of 5:  
Step 1: 1  
Step 2: 2  
Step 3: 6  
Step 4: 24  
Step 5: 120  
PS C:\Users\prath\.vscode\github> █
```

Question 3: Few-Shot Prompting (Armstrong Number Check)

Prompt : Write a Python program to check if a given number is an Armstrong number.

Please explain the logic using two examples in your explanation: **153** (which is an Armstrong number because $1^3 + 5^3 + 3^3 = 153$) and **123** (which is not an Armstrong number because $1^3 + 2^3 + 3^3 = 36$).

Code:

```
def is_armstrong(num):
```

```

# Convert the number to string to easily iterate over digits
str_num = str(num)
num_digits = len(str_num)

# Calculate the sum of each digit raised to the power of the number of digits
sum_of_powers = sum(int(digit) ** num_digits for digit in str_num)

# Check if the sum of powers is equal to the original number
return sum_of_powers == num

# Example usage
number1 = 153

if is_armstrong(number1):
    print(f"{number1} is an Armstrong number.")
else:
    print(f"{number1} is not an Armstrong number.")

number2 = 123

if is_armstrong(number2):
    print(f"{number2} is an Armstrong number.")
else:
    print(f"{number2} is not an Armstrong number.")

```

output:

```

PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"
153 is an Armstrong number.
123 is not an Armstrong number.
PS C:\Users\prath\.vscode\github> 

```

Question 4: Context-Managed Prompting (Optimized Number

Classification)

Prompt: to generate an optimized Python program that classifies a number as prime, composite, or neither.

Code:

```
def classify_number(num):  
    if num <= 1:  
        return "Neither prime nor composite"  
    elif num == 2:  
        return "Prime"  
    else:  
        for i in range(2, int(num**0.5) + 1):  
            if num % i == 0:  
                return "Composite"  
        return "Prime"  
  
# Example usage  
number = 29  
classification = classify_number(number)  
print(f"{number} is classified as: {classification}")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
29 is classified as: Prime  
PS C:\Users\prath\.vscode\github> █
```

Question 5: Zero-Shot Prompting (Perfect Number Check)

Prompt: generate a Python function that checks whether a given number is a perfect number.

Code:

```
def is_perfect_number(num):  
    if num <= 0:  
        return False  
  
    # Calculate the sum of proper divisors  
    sum_of_divisors = sum(i for i in range(1, num) if num % i == 0)  
  
    # A perfect number is equal to the sum of its proper divisors  
    return sum_of_divisors == num  
  
#example usage  
number = 28  
  
if is_perfect_number(number):  
    print(f"{number} is a perfect number.")  
else:  
    print(f"{number} is not a perfect number.")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
28 is a perfect number.  
PS C:\Users\prath\.vscode\github> █
```

Question 6: Few-Shot Prompting (Even or Odd Classification with Validation)

Prompt: Write a Python function to check whether a given number is odd or even. Please explain the logic using the **modulo operator (%)** and include two examples in your explanation: **4** (which is even because it is perfectly divisible by 2) and **7** (which is odd because it leaves a remainder of 1)

Code:

```
def is_odd_or_even(num):
```

```
if num % 2 == 0:  
    return "Even"  
else:  
    return "Odd"
```

Example usage

```
number1 = 4
```

```
number2 = 7
```

```
print(f"{number1} is {is_odd_or_even(number1)}.")
```

```
print(f"{number2} is {is_odd_or_even(number2)}.")
```

output:

```
PS C:\Users\prath\.vscode\github> python -u "c:\Users\prath\.vscode\github\tempCodeRunnerFile.py"  
4 is Even.  
7 is Odd.  
PS C:\Users\prath\.vscode\github> █
```